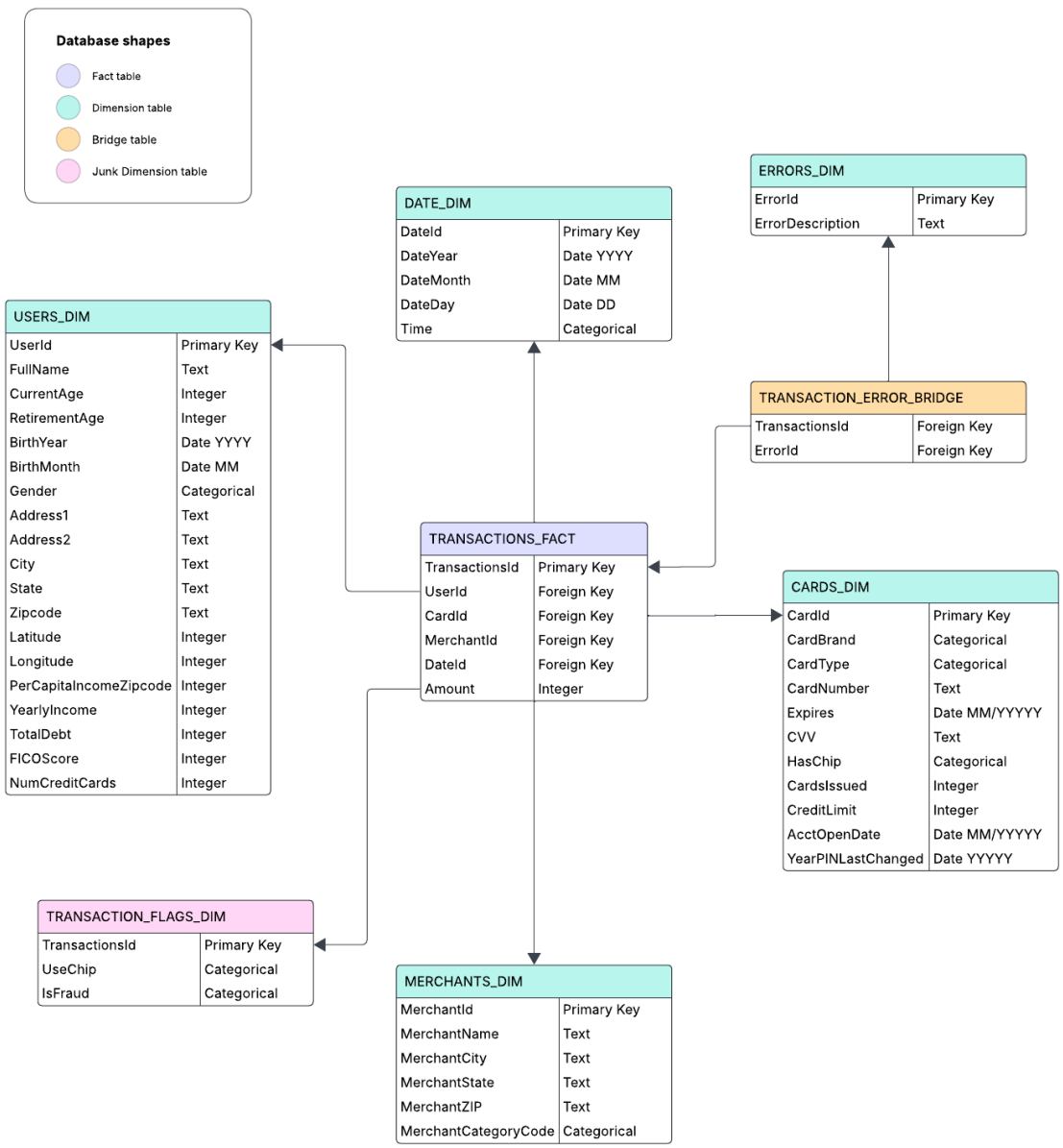
ANA 515 Project Part 2
Stage 1: Advanced Dimensional Modeling and Data Quality Assessment

<u>Title: Introducing Advanced Dimensional Modeling and Data Quality Assessment to the ETL pipeline for Credit Card Transactions Analysis</u>

Name: *Anushree Das*
Email: *anushree.das@mcdaniel.edu*

1. Updated Star Schema Design



Fact Table: Transactions

- Metrics: Amount
- Primary Key: TransactionsId
- Foreign Keys: UserId, CardId, DateId

Dimension Tables:

- Users Dimension: UserId, FullName, CurrentAge, RetirementAge, BirthYear, BirthMonth, Gender, Address1, Address2, City, State, Zipcode, Latitude, Longitude, PerCapitaIncomeZipcode, YearlyIncome, TotalDebt, FICOScore, NumCreditCards
- Cards Dimension: CardId, CardBrand, CardType, CardNumber, Expires, CVV, HasChip, CardsIssued, CreditLimit, AcctOpenDate, YearPINLastChanged
- Merchants Dimension: MerchantId, MerchantName, MerchantCity, MerchantState, MerchantZIP, MerchantCategoryCode
- Date Dimension: DateId, DateYear, DateMonth, DateDay, Time
- Errors Dimension: ErrorId, ErrorDescription

Junk Dimension Tables:

- Transaction Flags Dimension: UseChip, IsFraud

Bridge Table:

- Transaction Error Bridge: TransactionsId, ErrorId

2. ETL and Data Quality Documentation:

More detailed explanation of each step of the is provided in the R Code

Steps:

1. **Extract Data:**
    a. Imported CSV files for transactions, users, and cards.
2. **Transform Data:**
    a. Add column for primary key if not present
    b. Convert categorical columns using factor()
    c. Convert columns that are detected as Numeric but should be Text, for example, Zipcode
    d. Convert columns that are detected as Text but should be Numeric, for example, YearlyIncome
    e. Rename column names
    f. Remove duplicates
    g. Convert columns which has Dates to Date data type, for example, AcctOpenDate

h. Split Time column in the Transactions table into categories (morning, afternoon, evening, night)
i. Create more Dimension tables from Transactions table with columns with redundant data, for example, Merchant columns have same combination of values for multiple rows.

<u>New Steps Added to the Pipeline:</u>

Advanced Dimension Modeling

j. Create Junk Dimension table from Transactions table with flag columns.
k. Split column with list values and create Dimension table for it, eg, Errors column in the Transaction table.
l. Create bridge table to map the above table to the original

Data Quality Assessment

m. Identify missing values in all tables
n. Handle missing values for text column Address2 in Users table, with empty string to prevent NA values to affect the statistics for that column
o. Handle missing values by dropping rows in Merchants table where ZIP is NA, since it is an important feature for location based analysis of merchants and the count of rows with missing zip is very small(1267) when compared with the total number of rows(100,343) in the merchants table
p. Detect outliers in the metrics for the fact table transactions.
q. Transform column by categorizing the values in Amount column in Transactions table into "Low", "Medium", "High", “Extreme" categories.
r. Detect outliers in other numeric columns in Dimension tables and handle outliers using the IQR method
s. Standardize numeric variables to ensure consistent range across features using scale()
t. Convert text columns to lowercase and trim whitespace

3. **Load Data:**
   a. Created separate dataframes for fact and dimension tables in R.

3. Summary Report

- **Challenges and Solutions:** In refining our data warehouse for better analytics and modeling, I encountered several challenges that required careful design decisions and practical compromises to balance accuracy, performance, and interpretability.
  - To handle the columns which have complex values, for example, Errors column in Transactions table is a list of error types, I had to extract the list values into a dedicated

Errors dimension and create a bridge table to manage the many-to-many relationship between transactions and errors. While this greatly improved query accuracy and enabled precise error-based reporting, the transformation process itself was computationally expensive.

- Another major issue was handling outliers in the Amount column. The column had a high number of outliers on the upper end, ranging up to $12,000 while the third quartile was just around $65. Applying the interquartile range (IQR) method to detect outliers resulted in a large percentage of transactions being flagged, many of which were valid high-value transactions. Assigning the outliers with the upper bound value calculated using the IQR values resulted in uneven distribution, introducing a spike at the right end in the data distribution, To mitigate this, I converted the column into categorical bins (e.g., "Low", "Medium", "High", "Very High") to simplify the data and allow easier classification. While not ideal for all use cases, this approach will work well for fraud detection, where identifying patterns within specific transaction ranges is more valuable than knowing exact amounts. However, for use cases like predicting transaction amount or modeling user spending behavior, this transformation would reduce precision and potentially result in information loss.

- **Insights derived from the refined warehouse**
  - Normalizing multidimensional data (Errors in Transaction table) should allow us to accurately assess which types of errors frequently co-occurred and which were most predictive of fraud. It will also improved data governance by enforcing structure and consistency.
  - The transformation of the Amount column into categorical data will allow the fraud detection model to operate with better stability and less noise from extreme values, improving precision in classification tasks.